EASY ALGEBRA

Linear functions

01

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29

Q1 ALGEBRA

If f is the function defined by $f(x)=\frac{2x-1}{3}$, what is the value of $f(5)$?

A. $\frac{4}{3}$

B. $\frac{7}{3}$

C. 3

D. 9

Q2 GRID-IN ALGEBRA

The function g is defined by $gx = 6x$. For what value of x is $g(x) = 54$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GRID-IN ALGEBRA

The function f is defined by $fx = 4x$. For what value of x does $fx = 8$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4 ALGEBRA

Gabriella deposits $\$ 35$ in a savings account at the end of each week. At the beginning of the 1st week of a year there was $\$ 600$ in that savings account. How much money, in dollars, will be in the account at the end of the 4th week of that year?

A. 460

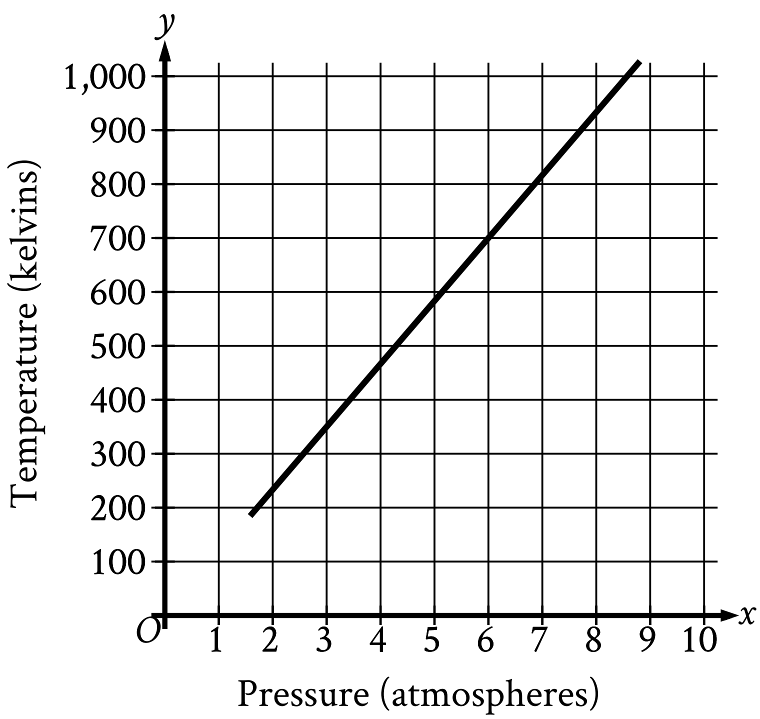
B. 635

C. 639

D. 740

Q5 ALGEBRA

Oxygen gas is placed inside a tank with a constant volume. The graph shows the estimated temperature y, in kelvins, of the oxygen gas when its pressure is x atmospheres.



- The line slants sharply up from left to right.
- The line begins at the approximate point (1.6 comma 190).
- The line passes through the following approximate points:
  - (3 comma 350)
  - (6 comma 700)
  - (8 comma 933)

What is the estimated temperature, in kelvins, of the oxygen gas when its pressure is 6 atmospheres?

A. 6

B. 60

C. 700

D. 760

Q6 ALGEBRA

The number y is 84 less than the number x. Which equation represents the relationship between x and y?

A. $y = x + 84$

B. $y = \frac{1}{84}x$

C. $y = 84x$

D. $y = x - 84$

Q7

ALGEBRA

x	$f(x)$
1	5
3	13
5	21

Some values of the linear function f are shown in the table above. Which of the following defines f ?

A. $f(x) = 2x + 3$

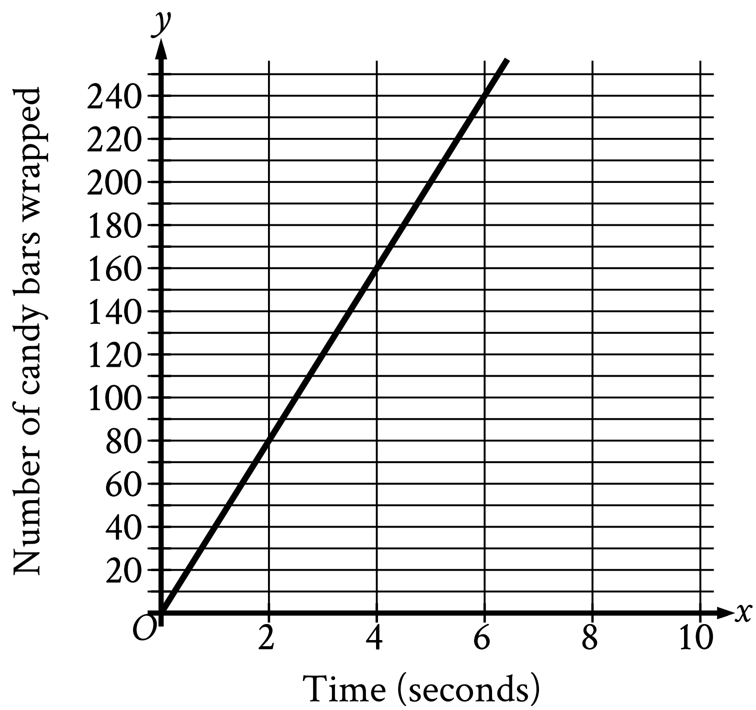
B. $f(x) = 3x + 2$

C. $f(x) = 4x + 1$

D. $f(x) = 5x$

Q8 ALGEBRA

The graph shown models the number of candy bars a certain machine wraps with a label in x seconds.



- The line slants sharply up from left to right.
- The line passes through the following points:
  - (0 comma 0)
  - (2 comma 80)
  - (4 comma 160)
  - (6 comma 240)

According to the graph, what is the estimated number of candy bars the machine wraps with a label per second?

A. 2

B. 40

C. 78

D. 80

Q9

ALGEBRA

The function f is defined by the equation $f x = 100x + 2$. What is the value of $f x$ when $x = 9$?

A. 111

B. 118

C. 900

D. 902

Q10 ALGEBRA

$$f(x) = 4x + b$$

For the linear function f, b is a constant and $f(7) = 28$. What is the value of b?

A. 0

B. 1

C. 4

D. 7